Element C

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Before researching this project, Dr. Brossman contacted our engineering teacher, Mr. Muscarella, about an e4usa project.

Dr. Brossman (Drexel University)
"I am an assistant professor at Drexel University teaching cardiovascular and pulmonary physical therapy. In this course we have 60-95 students annually. I teach about objective measures for respiratory evaluation. One objective measure we have is maximal inspiratory pressures or MIP. MIP is a measure of the strength of muscles for taking a breath in -or inhaling. The primary muscle of inhalation at rest is the diaphragm, and other muscles support the diaphragm when someone is maximally exercising or in a diseased state (sternocleidomastoid, trapezius ect).

MIP is of interest to medical providers as it can identify lung diseases that limit inhalation, ventilatory failure and respiratory muscle strength or respiratory muscle insufficiency.

To measure MIP a person generally inhales through a device, a manometer, which records the maximal amount of pressure that a person can generate. Normative values for MIP vary greatly in studies. We generally use MIP with the person as their own baseline and demonstrate changes over time when providing a strengthening routine to the diaphragm with an inspiratory muscle trainer or IMT. I generally like the Phillips threshold IMT. An IMT is a device that a person breathes through which has a pressure load that you can increase or decrease like weight training for the diaphragm. You measure IMT after a period of training and compared to a person's baseline MIP to see if you have changed strength over time.

We have a cohort of 97 this year, and have 1 (one) powerbreathe K series device to measure MIP (POWERbreathe K-Series IMT) and it was $2,000 and it is not economical to purchase enough devices for this class. I also only have a few Phillips Threshold IMT and they are $100.00 and neither are covered by insurance. I want to allow ALL my students to measure their own MIP in class, and to experience IMT as we know this translates to use of this measurement for their patients as they move to become excellent physical therapists.

The limiters/barriers in terms of cost for these devices often impact those who need them the most as they are quite expensive. During COVID many more underrepresented minorities died due to respiratory dysfunction. I believe the algorithms used to measure respiratory impairment were flawed or not used. Some of these when not measured are calculated with flawed metrics based on race. I want students to measure directly and not use a flawed calculation. I purchased a negative inspiratory force (NIF) meter (Negative Inspiratory Force (NIF) Meters) which is also a few hundred dollars.

What I need: a safe (able to be cleaned and shared between students), inexpensive, reliable and valid way to measure MIP with enough devices to be able to support a large cohort in minimal time. For our cohort of 97 students the lab is an hour during which they rotate into a MIP measuring station but we need enough devices for 66a number of students (around 10-20) to use at one."

Design Requirements			
Rankings	Requirements	Why	Research behind it
1 (High Priority)	Safe	Being safe to use is the number one priority for users, considering that this device is for medical purposes. Safety must be ensured by preventing infections from the device due to excessive use of the mouthpiece. The MIP should be made with any sharp edges covered that the users could potentially harm themselves on.	Research shows that after a breathing device is used, it will be exposed to saliva and possible infections. The problem that needs to be solved for the MIP is being able to clean the device after every use. https://www.pnmedical.com/Cleaning/

2 (High Priority)	User-friendly	The students must be able to use the device easily. If the device is to be mass-produced and distributed to people in need, they must be able to use it with little instruction or help. “During COVID, many more underrepresented minorities died due to respiratory dysfunction. I believe the algorithms used to measure respiratory impairment were flawed or not used. Some of these, when not measured, are calculated with flawed metrics based on race.”	Dr. Brossman’s letter
3 (High Priority)	Precise	The device's goal is to measure how strong one can inhale. It would not be ideal if it could not provide exact or pinpoint results.	Respir Care. Author manuscript; available in PMC: 2013 Apr 4. Published in final edited form as: Respir Care. 2009 Oct;54(10):1321–1328. https://pmc.ncbi.nlm.nih.gov/articles/PMC3616895/ Respiratory muscle weakness is a predictor of all-cause mortality and MIP has previously been shown to be associated with incident cardiovascular events including myocardial infarction, cardiovascular death and possibly stroke. MIP allows for the assessment of ventilatory

			failure, restrictive lung disease and respiratory muscle strength. The device would need to be precise in order to determine if a user has all of those factors.
4 (High Priority)	Repeatability	The device must be able to have students consistently change their level of endurance. This would be a top priority as the device's main purpose is to have patients improve their respiratory endurance. In addition, as the device will be used multiple times by cohorts 20 students, the object should be able to endure multiple trials	During research and customer reviews, patients enjoy the fact that the Philips Threshold IMT has adjustable resistance as this improved their respiratory endurance.
5 (High Priority)	Inexpensive	Many MIP devices are very expensive ($500+). For the purpose of Dr. Brossman's class, she needs multiple devices for larger groups of students. Manufacturing the device in a simple/ inexpensive manner will provide a cheap solution she can buy in bulk.	Source After researching different kinds of MIP devices, most range from $200- $700+ dollars depending on how technical the device is. https://www.powerbreathe.com/us/product-category/breathing-trainers-imt-and-emt/k-series-imt/
6 (Low Priority)	Accuracy	The goal of the device is to see how strong one can inhale. It would not be ideal if the device provided inaccurate results	Respir Care. Author manuscript; available in PMC: 2013 Apr 4. Published in final edited form as: Respir Care. 2009 Oct;54(10):1321–1328. https://pmc.ncbi.nlm.nih.gov/articles/PMC3616895/ Respiratory muscle weakness is a predictor of

			all-cause mortality and MIP has previously been shown to be associated with incident cardiovascular events including myocardial infarction, cardiovascular death and possibly stroke. MIP allows for the assessment of ventilatory failure, restrictive lung disease and respiratory muscle strength. The device would need to be precise in order to determine if a user has all of those factors.
7 (Low Priority)	Small	Dr. Brossman did not consider size an issue. Based on the letter, the requirements did not include anything about size. Her email also said that size wasn't a huge issue since the device would stay in the lab.	Throughout researching for the device, all of the sources did not include size as a problem. It is implicit that these devices are supposed to be small and easy to work with. **Dr. Brossmans letter** does not stress the importance of the size of the MIP respiratory device.
8 (Low Priority)	Portable	This device is for multi-functioning purposes. The MIP device needs to be able to move between 10 and 20 students at a time. Portability is considered low quality because any device we make will be small.	Research shows that all different designs for this device can be transported easily. **Dr. Brossman's letter** does not stress the importance of the respiratory device's portability.
9 (Low Priority)	Aesthetic	The device's aesthetic is low priority since Dr. Brossman doesn't mention it at all in the letter. However, we can assume that Dr. Brossman does not want an	From research, no sources mentioned the look of any IMT device. However, we assume customers would want to spend their money on an aesthetically pleasing

		aesthetically unpleasing device.	device. **Dr. Brossmans letter** does not stress the importance of the aesthetics of the MIP respiratory device.